

askADZA Commercialization Cost Summary

Production cost model for the coming quarter — itemized per request, per month, and per quarter

Prepared for: OpenTrace / askADZA commercialization planning

Traffic basis: 5,000 production API requests/day (150,000/month · 450,000/quarter)

Price snapshot: August 2026 public list prices (OpenRouter, Railway, Qdrant, BigQuery, rerank, Tavily)

Document type: Planning estimate — not an invoice. Confirm live vendor routes before locking budget.

1. Executive headlines

Two planning headlines are provided. Scenario A is today's RAG graph. Scenario B adds DeepSeek R1 reasoner CoT for the next RAG update.

Scenario	\$ / request	\$ / month	\$ / quarter	Notes
A — Max product COGS (today)	~\$0.041	~\$6.2k	~\$18.5k	Integrated + full RAG + pessimistic BQ; Tavily off; excl. Streamlit ops
B — Max + DeepSeek R1 CoT	~\$0.051	~\$7.7k	~\$23.0k	Scenario A + ~\$0.01/req reasoner CoT (can be higher with long thinking)
Blended (illustrative mix)	~\$0.012–0.022	~\$1.8–3.3k	~\$5.4–9.9k	50% Free / 20% Farmers / 15% Gov+NGO / 15% Agri+Integrated; BQ on ~40%
Streamlit ops (extra)	—	~\$25	~\$75	Dev/QA only — not in product \$/request

Practical takeaway: Budget ~\$6–8k/month for worst-case Integrated + pessimistic BQ; add ~\$1.5k/month if every request also runs R1. Free-heavy mix with disciplined BQ is closer to \$2–3.5k/month.

2. Scope and reading guide

2.1 What is included

- Railway production API + Redis serving Ask ADZA user traffic
- Qdrant Cloud sized for ≥ 1 M vectors with HNSW $m=16$, plus ColBERT-on-disk and a ColPali pilot uplift

- Langfuse observability (self-hosted assumption)
- BigQuery analysis from RAG NL->SQL (up to 10 jobs per query)
- OpenRouter LLM calls (paid routes only — no free/:free models)
- API rerank (Cohere/OpenRouter rerank-4-pro at \$0.0025/search)
- Optional Tavily for paid plans only (default off in base max)

2.2 What is excluded or treated separately

- Streamlit — internal development / QA inspector only; does NOT receive 5k req/day; fixed ops line only
- Tavily on Free plan — not available; Free always \$0 for Tavily
- Frontend hosting, domains, auth/billing gateway, legal counsel, human ops/on-call
- Corpus ingestion / reindex GPU time (especially ColPali/ColBERT embedding at ingest)
- News-mining Tavily usage outside the chat path

2.3 What max means

Max = one Integrated-plan user query on the full RAG path: decompose -> hybrid retrieve -> up to 10 NL->SQL LLM calls -> up to 10 BigQuery jobs -> API rerank -> generate with Qwen3-235B. BigQuery assumes ~0.5 GiB scanned per job (5 GiB total). Upper-bound product COGS for capacity/abuse planning, not average-mix forecast.

3. Model ladder (generate)

Pipeline nodes (decompose, NL->SQL) use Llama 3.1 8B. Top tiers use Qwen only (no GLM).

Plan	Generate model	Tavily eligible?	Est. \$/1M tokens (in / out)
Free	meta-llama/llama-3.1-8b-instruct	No	\$0.02 / \$0.04
Farmers	meta-llama/llama-3.1-70b-instruct	Yes, if web fallback on	\$0.40 / \$0.40
Government	qwen/qwen-2.5-72b-instruct	Yes, if web fallback on	\$0.36 / \$0.40
NGOs	qwen/qwen-2.5-72b-instruct	Yes, if web fallback on	\$0.36 / \$0.40
Agribusinesses	qwen/qwen3-235b-a22b-instruct-2507	Yes, if web fallback on	~\$0.09 / \$0.55 (max route)
Integrated	qwen/qwen3-235b-a22b-instruct-2507	Yes, if web fallback on	~\$0.09 / \$0.55 (max route)

Deploy note: set RAG_LLM_MODEL_* env vars to this ladder — code defaults in plan_policy.py still differ.

4. What one max query burns

1. 1x decompose (Llama 3.1 8B)

2. Parallel Qdrant hybrid retrieve (local E5-small + BM25 — \$0 API)
3. Up to 10x NL->SQL (Llama 3.1 8B) and up to 10 BigQuery jobs
4. 1x rerank (API \$0.0025, or \$0 if local cross-encoder)
5. 1x generate (Qwen3-235B for Integrated / Agribusinesses)
6. Tavily: off by default; if on, paid plans only

5. Itemized costs — Scenario A (base max, no CoT)

5.1 Variable COGS

Line item	\$ / request	\$ / month (150k)	\$ / quarter (450k)	Basis
LLM pipeline (Llama 3.1 8B)	~\$0.00055	~\$83	~\$248	decompose + <=10 NL2SQL @ \$0.02/\$0.04 per 1M
LLM generate (Qwen3-235B)	~\$0.0012	~\$180	~\$540	~4k in / 1.5k out @ \$0.09/\$0.55 per 1M
Rerank (rerank-4-pro)	\$0.0025	~\$375	~\$1,125	1 search per query
BigQuery (pessimistic)	~\$0.0305	~\$4,575	~\$13,725	10 x 0.5 GiB = 5 GiB @ \$6.25/TiB
Tavily	\$0	\$0	\$0	Default off; Free never
Embeddings API	\$0	\$0	\$0	Local multilingual-e5-small
Variable subtotal	~\$0.0348	~\$5,213	~\$15,638	

BQ cliff: If scans average 50 MiB/job instead of 500 MiB, BQ falls to ~\$0.003/request (~\$450/month) and Scenario A all-in drops to roughly ~\$0.014/request.

5.2 Fixed / allocated infrastructure (production)

Fixed lines allocated as $\text{monthly_cost} / 150,000$. Streamlit excluded from product COGS.

Line item	\$ / month	\$ / request	\$ / quarter	Notes
Railway API + Redis	~\$65	~\$0.00043	~\$195	Serves 5k/day user traffic
Qdrant (HNSW m=16, >=1M + ColBERT disk)	~\$700	~\$0.0047	~\$2,100	Largest fixed line; confirm on Qdrant calculator
ColPali pilot uplift	~\$100	~\$0.00067	~\$300	Subset corpus only
Langfuse self-host	~\$75	~\$0.0005	~\$225	Prefer vs Cloud unit fees at this volume
GCS / egress	~\$30	~\$0.0002	~\$90	Exports + outbound
Prod fixed subtotal	~\$970	~\$0.0065	~\$2,910	

5.3 Streamlit (ops only)

Internal pipeline inspector. Keep private. Not part of 5k/day denominator.

Line item	\$ / month	\$ / quarter
Railway Streamlit (dev/QA, >=1 GB)	~\$25	~\$75

5.4 Scenario A totals

	\$ / request	\$ / month	\$ / quarter
Variable	~\$0.0348	~\$5.2k	~\$15.6k
Prod fixed allocated	~\$0.0065	~\$1.0k	~\$2.9k
Product total (Scenario A)	~\$0.041	~\$6.2k	~\$18.5k
+ Streamlit ops	—	+\$25	+\$75
Cash out (product + Streamlit)	—	~\$6.25k	~\$18.6k

6. Scenario B — alternative with reasoner CoT (DeepSeek R1)

Next RAG update adds DeepSeek R1 reasoner nodes. Reasoning models bill for long chain-of-thought completion tokens (often 2-4x a normal answer). NOT in Scenario A because nodes are not in today's production graph.

6.1 Reasoner cost assumption

Item	Assumption
Model	deepseek/deepseek-r1-0528 (or current R1 route on OpenRouter)
List price (approx.)	~\$0.50 / \$2.15 per 1M input / output tokens
Token budget per reasoner call (planning)	~3k input / ~4k output (includes CoT)
Calls per query (base alternative)	1 reasoner call
Est. \$/request	~\$0.01
Long-CoT risk	Can rise well above \$0.01 if thinking traces are long; \$0.01 is a floor for on, not a hard cap

Illustrative calculation: $(3,000/1e6) \times \$0.50 + (4,000/1e6) \times \$2.15 \approx \$0.0015 + \$0.0086 \approx \$0.0101$ per request.

6.2 Scenario B totals (A + one R1 CoT call)

	\$ / request	\$ / month (150k)	\$ / quarter (450k)
Scenario A product total	~\$0.041	~\$6.2k	~\$18.5k
+ DeepSeek R1 CoT	~+\$0.010	~+\$1.5k	~+\$4.5k
Scenario B product total	~\$0.051	~\$7.7k	~\$23.0k
+ Streamlit ops	—	+\$25	+\$75
Cash out (B + Streamlit)	—	~\$7.75k	~\$23.1k

If reasoner runs only on paid tiers or hard questions, multiply +\$0.01 by that traffic share instead of 100% of 5k/day.

7. Tavily (paid plans only)

- Free plan: always \$0 — gate in product/auth, not only marketing copy.
- Paid plans: eligible when RAG web fallback is enabled.
- Advanced search (~2 credits) at ~\$0.008/credit → about +\$0.016 per paid request that hits Tavily.
- Scenarios A and B assume Tavily off.
- News-mining / ingestion Tavily is a separate ops budget.

8. Blended planning mix (illustrative)

50% Free, 20% Farmers, 15% Gov+NGO (Qwen2.5-72B), 15% Agri+Integrated (Qwen3-235B); BQ on ~40%; rerank on; Tavily off; no R1.

Metric	Band
\$ / request	~\$0.012 – \$0.022
\$ / month	~\$1.8k – \$3.3k
\$ / quarter	~\$5.4k – \$9.9k

Adding R1 on 15% of traffic (top tiers only) $\approx 0.15 \times \$0.01 = \sim \$0.0015/\text{request}$ overall ($\sim \$225/\text{month}$).

9. Sensitivities

Change	Delta \$ / request	Comment
DeepSeek R1 reasoner (1 call)	+~\$0.01	Scenario B; higher if CoT is long
NL->SQL on Llama 70B instead of 8B	+~\$0.07	Misconfig / abuse cliff
BQ 50 MiB/job vs 500 MiB	-~\$0.027	Biggest controllable saving
Local cross-encoder rerank	-\$0.0025	CPU on Railway instead of API
Tavily on 100% of paid requests	+\$0.016 per paid req	Free remains \$0
Qwen3-Max instead of Qwen3-235B	generate toward ~\$0.009	At ~\$0.78/\$3.90 per 1M

10. Vendor narratives

10.1 Railway

At 5k/day, Railway is a small share of COGS. Split production API+Redis from Streamlit QA. Do not scale Streamlit as user traffic.

10.2 Qdrant

Largest fixed commitment. HNSW m=16 plus $\geq 1M$ hybrid points suggests ~16-32 GB Standard (~\$450-900/month; midpoint \$700) plus ColBERT disk and ColPali pilot. Confirm on <https://cloud.qdrant.io/calculator>.

10.3 Langfuse

Full tracing at 5k/day creates millions of Cloud billable units/month. Model assumes self-hosted OSS (~\$75/month).

10.4 BigQuery

Dominant variable risk. Billing is bytes scanned, not rows returned. Enforce filters, partitions, and `maximumBytesBilled`.

10.5 OpenRouter LLMs

Qwen3-235B keeps top-tier generate inexpensive. Real cliffs: 70B on all NL2SQL calls, and R1 CoT token blowups.

11. Public launch gaps that affect cost

Not inside \$0.041 / \$0.051, but they decide whether those figures survive real users:

1. API authentication — currently none; open URL can burn OpenRouter/BQ/Qdrant
2. Plan enforcement — client-supplied `plan_type` can be spoofed; Free must not get Integrated models or Tavily
3. Rate limits (Redis-backed) and OpenRouter / BQ spend caps
4. CORS lockdown; protect `/docs`; keep Streamlit private
5. BQ dataset `raw_prod` (not `raw_dev`); GCS for paid exports
6. Scheduled ingestion / Qdrant backup; empty OTA messaging until corpus is loaded
7. Legal / retention for chat in Redis and Langfuse
8. Frontend billing, domains, staging (PR deploys can double Railway spend)

12. Recommendations

- Use Scenario A (~\$6.2k/month max) as near-term production budget without reasoners.
- Hold Scenario B contingency (~+\$1.5k/month at 100% R1, or pro-rate by tier) before enabling CoT reasoners.
- Prioritize BigQuery scan controls — they move the number more than generate model choice.
- Ship auth + plan gates + spend caps before public URL.
- Keep Streamlit private and out of product COGS reporting.

- Re-check OpenRouter route prices and Qdrant calculator before board approval.

13. Disclaimer

All figures are planning estimates from public pricing and the askADZA/RAG architecture as of the document date. Vendor prices and routing change. Validate with dry-run BigQuery bytes, OpenRouter invoices, and the Qdrant Cloud calculator before committing spend.